



*In memoriam Max Klein
(1951-2024)*

High-energy two-photon interactions at future ep colliders

DIS 2025, Cape Town, South Africa

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Multiple tracks for **(polarised/unpolarised) electron-hadron machines** in the 21st Century, incl. some presented at this workshop:

■ EIC @ BNL [1, 2, 3]

10 – 18 GeV (e^-) $\oplus$ $\mathcal{O}(100 \text{ GeV})$ (A) ($\sqrt{s} \sim 20 - 140 \text{ GeV}$) $\Rightarrow$ 2030s

■ LHeC + FCC-eh @ CERN [4, 5, 6]

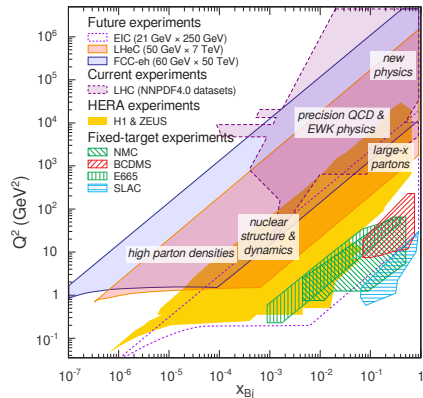
50 GeV (e^-) $\oplus$ 7 TeV (p) for LHeC ($\sqrt{s} \sim 1.3 \text{ TeV}$) $\Rightarrow$ 2040-2050s

20 GeV (e^-) — @ **phase-1 LHeC** ($\sqrt{s} \sim 0.75 \text{ TeV}$) 

arXiv:2503.20475

60 GeV (e^-) $\oplus$ 100 TeV (p) for FCC-eh ($\sqrt{s} \sim 3.5 \text{ TeV}$)

■ CEPC-SehC



In particular, LHeC is expected to deliver an integrated e-p luminosity of $\sim 1 \text{ ab}^{-1}$ over its lifespan

Rich playground for high precision **electroweak & QCD** studies at the TeV scale

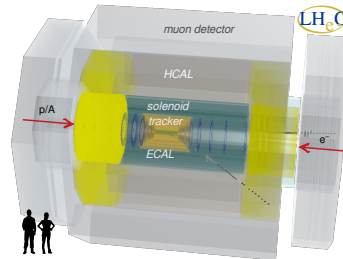
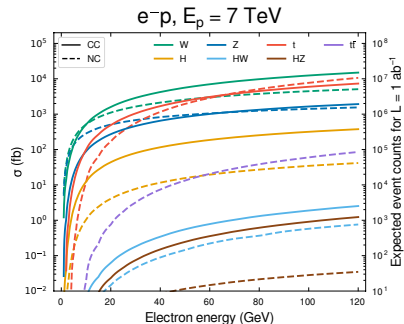
- best limits on m_W , m_Z , aTGCs
- studies of top quark properties (coupling, mass, ...) through DIS/FCNCs/diffractive production processes
- DIS CC/NC Higgs production to study its characteristics
- extensive DIS/QCD programme at broad x_{Bj} , or high- Q^2 values

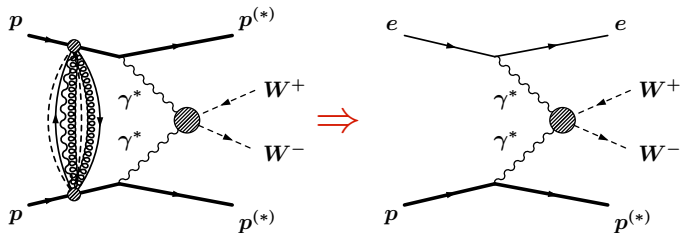
Much more than a super-HERA collider

- $\mathcal{L}_{\text{LHeC}} > 10^3 \mathcal{L}_{\text{HERA}}$
- high discovery potential for new physics
- can also operate in electron-ion mode

More details about main LHeC prospects: see N. Armesto's [7, 8], C. Gwenlan's [9], and K. Wichmann's [10] talks

⊕  **"LHeC as a bridge"** white paper arXiv:2503.17727 [hep-ex]





Given the **good sensitivity** shown by $\gamma\gamma$ searches at LHC, transposition to the e-h / LHeC case simplifies the problem:

- very low event pileup, ensuring a very **clean final state**
- barely any hadronic re-scattering activities spoiling the rapidity gap between the central system & forward scattered beam particles
- beam-beam interaction largely reduced, mostly soft corrections from ISR

Factorisation of the **central, two-photon process matrix element** from the **total e-p cross section**:

$$\begin{aligned}\sigma(ep \rightarrow e(\gamma\gamma \rightarrow X)p^{(*)}) &= \int dy_e \Phi_e(y_e) \int dy_p \Phi_{p^{(*)}}(y_p) \sigma_{\gamma\gamma \rightarrow X}(W \equiv \sqrt{s y_e y_p}) \\ &= \int dW S_{\gamma\gamma}(W) \sigma(W),\end{aligned}$$

where the $\gamma\gamma$ **luminosity spectrum** $S_{\gamma\gamma}$ is the convolution of both photon-from-electron, and photon-from-proton fluxes:

$$S_{\gamma\gamma}(W) = \frac{2}{W} \int_{W^2/s}^1 dy_e \Phi_e(y_e) y_p \Phi_p(y_p) = \frac{2W}{s} \int_{W^2/s}^1 dy_e \frac{\Phi_e(y_e)}{y_e} \Phi_p\left(\frac{W^2}{y_e s}\right),$$

with **form factors**-dependent photon fluxes:

$$\Phi(y) = \frac{\alpha_{em}}{\pi} \frac{1}{y} \int \frac{dQ^2}{Q^2} \left\{ (1-y) \left(1 - \frac{Q_{min}^2}{Q^2} \right) F_E(Q^2) + \frac{y^2}{2} F_M(Q^2) \right\}$$

- **Photon-from-electron** → trivial form factors: $F_M = F_E = 1$
- **Photon-from-proton**: 2 types of γ emission: elastic / dissociative (w/ proton fragmentation)
 - Elastic scattering: **Sachs form factors**-dependent:

$$F_E(Q^2) = \frac{4m_p^2 G_E^2 + Q^2 G_M^2}{4m_p^2 + Q^2}, \quad \text{and} \quad F_M(Q^2) = G_M^2$$

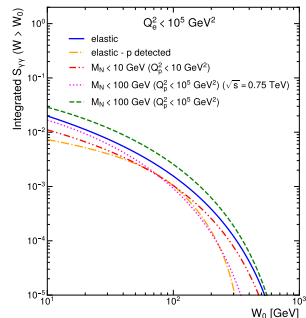
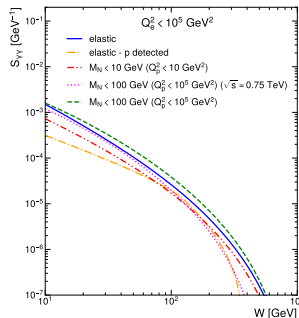
- Inelastic scattering: **nucleon structure functions**-dependent ; at first order, only F_2 -dependent:

$$F_E(Q^2) = \int \frac{dx}{x} F_2(x, Q^2), \quad \text{and} \quad F_M(Q^2) = \int \frac{dx}{x^3} F_2(x, Q^2)$$

E.g., for the **LHeC** energy range:

- ALLM F_2 parameterisation
 $(10^{-6} < x_{\text{Bj}} < 0.85, Q^2 < 5000 \text{ GeV}^2)$
 for inelastic scattering

⇒ central system kinematics reach
 limited by **behaviour at high-mass**
 of $S_{\gamma\gamma}$



Other approaches for $\gamma\gamma \rightarrow X$ predictions

According to the process, other techniques can be implemented for a more precise prediction of $\gamma\gamma$ processes

E.g. “HERA” Monte Carlo generators for $\gamma^{(*)}\gamma \rightarrow \ell^+\ell^-$:

LPAIR¹

- **pros**: full matrix element computation, suitable for e-p as for p-p (using electromagnetic form factors/DIS structure functions)
- **cons**: only **Bethe-Heitler** production implemented, no ISR/EW corrections, Suri & Yennie as only continuum SF

⇒ now implemented in CepGen²

GRAPE³

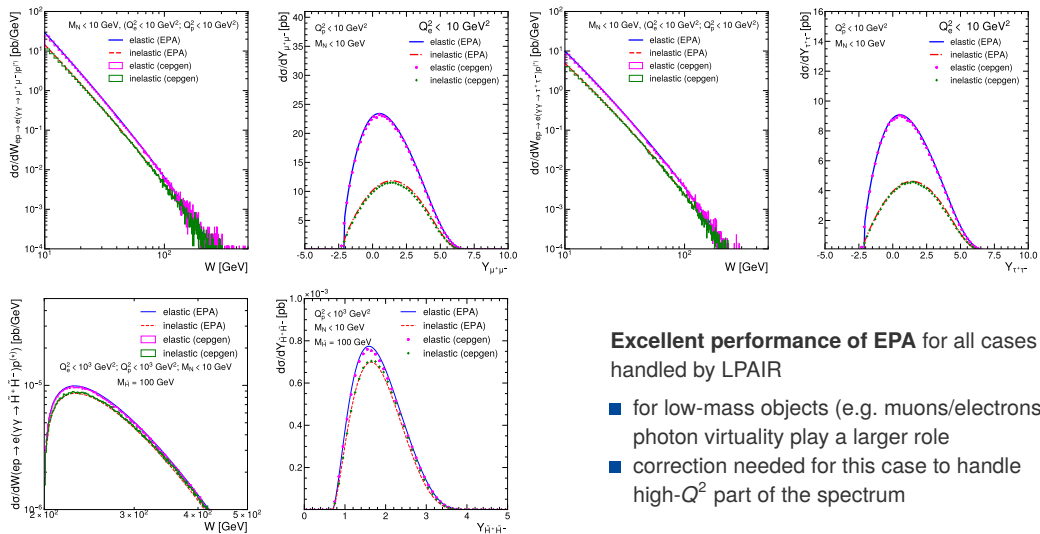
- **pros**: same approach as LPAIR (full matrix element computation), with all helicity amplitudes ; also includes **ISR/EW corrections** (incl. Z exchanges) ; accounts for **beam polarisation**
- **cons**: severely outdated stack, very slow and numerically unstable for corner cases of the phase space

⇒ implementation in CepGen under validation

¹Nucl.Phys.B 229 (1983) 347

²Comput.Phys.Comm. 271 (2022) 108225, arXiv:1808.06059 [hep-ph]

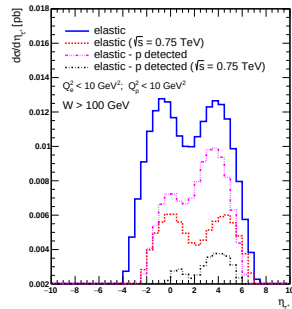
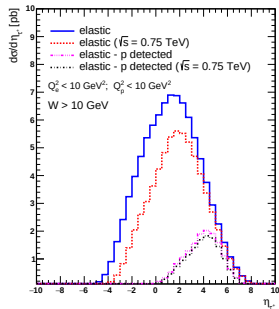
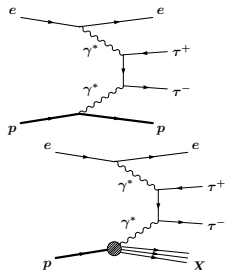
³Comput.Phys.Comm. 136 (2001) 126, arXiv:hep-ph/0012029



Excellent performance of EPA for all cases handled by LPAIR

- for low-mass objects (e.g. muons/electrons), photon virtuality play a larger role
- correction needed for this case to handle high- Q^2 part of the spectrum

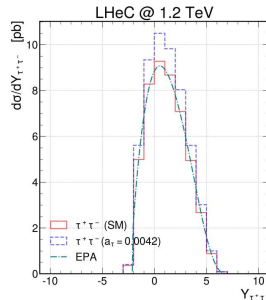
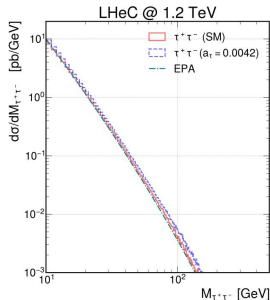
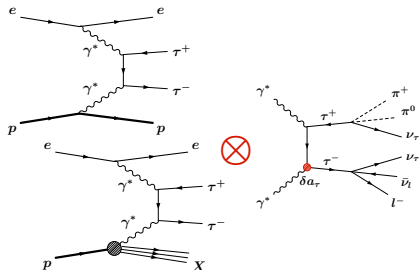
Physics case 1: $\gamma\gamma \rightarrow \tau^+\tau^-$



LHeC (1 ab^{-1}): $\mathcal{O}(10^8)$ $\gamma\gamma \rightarrow \tau^+\tau^-$ events produced at $W > 10 \text{ GeV}$, $\sim 2 \times 10^6$ at $W > 50 \text{ GeV}$

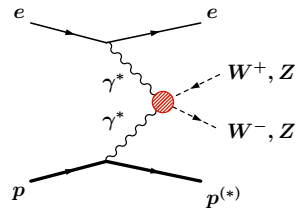
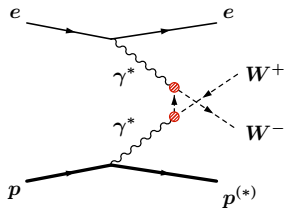
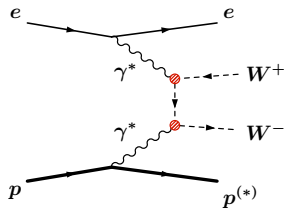
- magnitude of ISR corrections at the order of 5
- main background from other lepton flavours in $\gamma\gamma \rightarrow \ell^+\ell^-$ (sensitivity enhanced for **semi-leptonic decay modes**), $\gamma\gamma \rightarrow W^+W^-$, $\gamma\gamma \rightarrow q\bar{q}$, NC DIS, photoproduction processes ...

Physics case 1: $\gamma\gamma \rightarrow \tau^+\tau^-$



LHeC (1 ab⁻¹): $\mathcal{O}(10^8)$ $\gamma\gamma \rightarrow \tau^+\tau^-$ events produced at $W > 10$ GeV, $\sim 2 \times 10^6$ at $W > 50$ GeV

- magnitude of ISR corrections at the order of 5
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- allows to probe with high precision the τ/γ coupling, sensitive to anomalous **SMEFT deviations** of electromagnetic moment $\delta(a_\tau \equiv (g_\tau - 2)/2)$ [Rep. Prog. Phys. 87 (2024) 107801, arXiv:2503.19836 [hep-ex]]



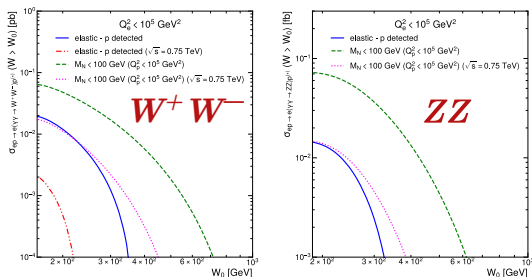
In the **high energy limit**, $\sigma_{\gamma\gamma}(W) \propto \begin{cases} c & \text{for vector particles} \\ 1/W & \text{for fermions} \\ 1/W^2 & \text{for scalar particles} \end{cases} \Rightarrow \text{powerful test bench for the probing of anomalous triple/quartic gauge couplings}$

$\gamma\gamma \rightarrow W^+W^-$ already broadly studied in
LHC phase space e.g. [CMS: JHEP 07 (2023) 229,
ATLAS: Phys.Lett.B 816 (2021) 136190]

- very challenging under increasingly difficult conditions

LHeC cleanliness allowing **precision probe**
of γVV & $\gamma\gamma VV$ couplings

- as for CEP $\tau^+\tau^-$ production, multiple (fully leptonic, semileptonic, fully hadronic) decay modes can be considered
- **EFT models** predicting anomalous production processes (yield + differential kinematics distributions)
- in SM, $\gamma\gamma \rightarrow ZZ$ not allowed at tree level: ≥ 1 loop involved & $\sigma_{\gamma\gamma \rightarrow ZZ}^{\text{SM}} \ll$



$$ep \rightarrow e(\gamma\gamma \rightarrow W^+W^-)p^{(*)}$$

$\sigma_{\text{el}} \text{ (fb)}$	$\sigma_{\text{inel}} \text{ (fb)}$		
	$M_N < 10 \text{ GeV}$	$M_N < 50 \text{ GeV}$	$M_N < 300 \text{ GeV}$
33.4	29.3	56.5	70.8

Broad physics case for $\gamma\gamma$ physics at electron-hadron colliders

In particular LHeC offers unique conditions for studying high energy photon-photon interactions:

- very low event pileup and data streaming allow to measure two-photon exclusive production with high efficiency ;
 - addition of forward proton/shower detectors for exclusivity selection is a plus
- large photon-photon luminosities makes it an ideal laboratory for $\gamma\gamma$ physics ;
- excellent prospects for the study of exclusive two-photon production of lepton pairs, pairs of W and Z bosons, and for other searches (e.g. charged supersymmetric/BSM particles).

High energy $\gamma\gamma$ physics at LHeC will also significantly **enhance LHeC scientific potential**

- with accelerated (bridge) LHeC, will complement HL-LHC scientific endeavours, particularly in QCD & electroweak sectors
- staged, $E_e = 20$ GeV also very competitive for $\gamma\gamma$ searches

Thanks for your attention!

Spares

High energy two-photon processes provide unique opportunity for studying with high precision the electroweak sector of particle interactions at future colliders.

In particular, future facilities such as the LHeC, FCC-eh, or the eh-CEPC, will ensure very advantageous experimental conditions and at the same time provide high two-photon luminosities, reaching the TeV scale.

In this talk we will give an overview of expected performance of these colliders in that context. In addition, the impact of tagging two-photon processes using very forward detectors will also be presented.